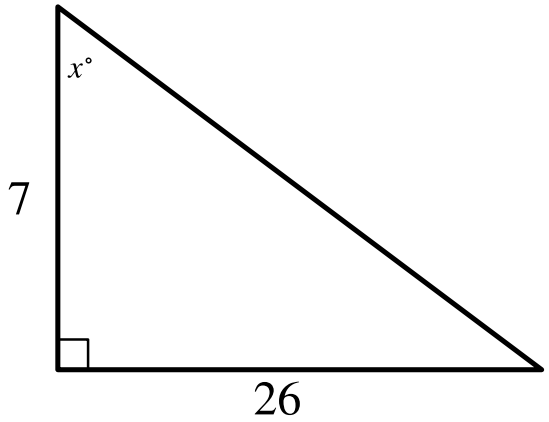
Right triangles and trigonometry

Geometry and Trigonometry · Math · 69 questions

1 359c6a4b-24df-4f06-8a59-45dbbbb00833 · Easy



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the leg opposite the angle with measure x° is 26.
- The length of the leg adjacent to the angle with measure x° is 7.
- A note indicates the figure is not drawn to scale.

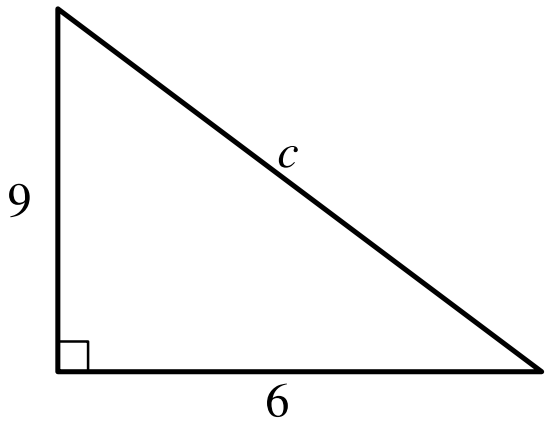
In the triangle shown, what is the value of $\tan x°$?

A. $\frac{1}{26}$

B. $\frac{19}{26}$

C. $\frac{26}{7}$

D. $\frac{33}{7}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is c.
- The lengths of the other 2 sides are as follows:
  - 9
  - 6
- A note indicates the figure is not drawn to scale.

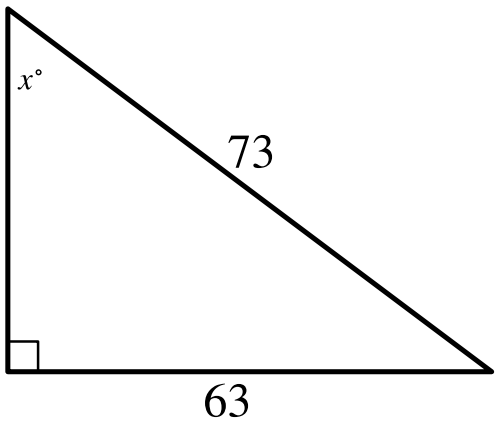
In the right triangle shown, which of the following is closest to the value of *c*?

A. 7.5

B. 10.8

C. 15

D. 58.5



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the side opposite the right angle is 73.
- The length of the side opposite the angle with measure x° is 63.
- A note indicates the figure is not drawn to scale.

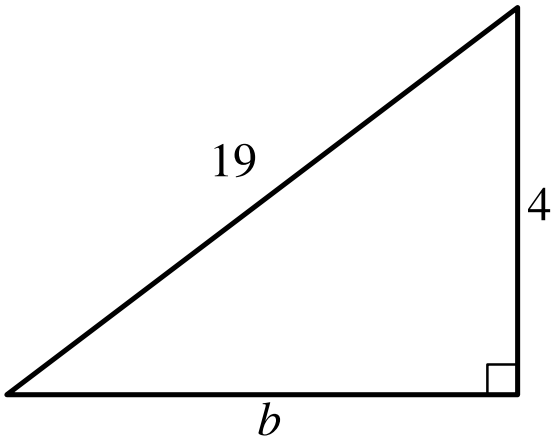
In the right triangle shown, what is the value of $\sin x°$?

A. $\frac{1}{73}$

B. $\frac{10}{73}$

C. $\frac{63}{73}$

D. $\frac{136}{73}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 19.
- The lengths of the other 2 sides are as follows:
  - b
  - 4
- A note indicates the figure is not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

A. $4b = 19$

B. $4 + b = 19$

C. $4^2 + b^2 = 19^2$

D. $4^2 - b^2 = 19^2$

5 924b44e4-dcc1-47ff-ad3e-564a5a1991ef · Easy

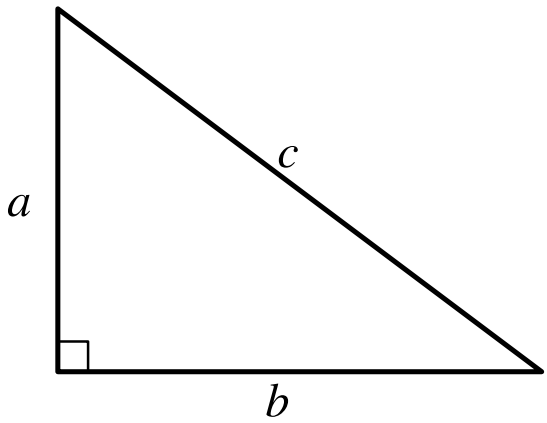
Triangle *ABC* is similar to triangle *DEF*, where angle *A* corresponds to angle *D*, and angles *C* and *F* are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

A. $\frac{1}{22}$

B. $\frac{1}{23}$

C. $\frac{21}{22}$

D. $\frac{22}{23}$



Note: Figure not drawn to scale.

- The side lengths are labeled as follows:
  - a
  - b
  - c
- The angle opposite the side labeled c is a right angle.
- A note indicates the figure is not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c?

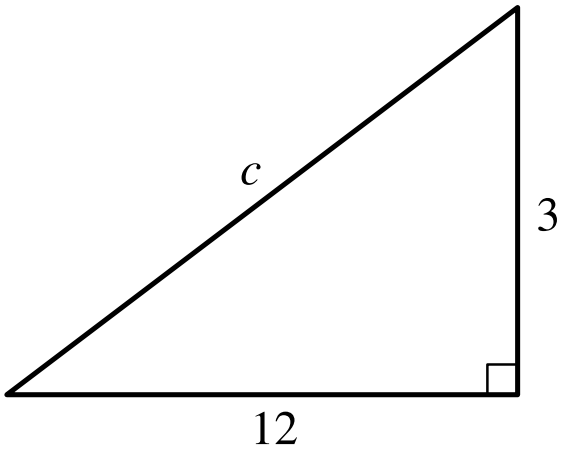
A. $4 + 5$

B. $\sqrt{45}$

C. $\sqrt{4 + 5}$

D. $\sqrt{4^2 + 5^2}$

7 aefa5414-12c1-4240-9455-47ff78e9896d · Easy



Note: . igure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is c.
- The lengths of the other 2 sides are as follows:
  - 3
  - 12
- Note: Figure not drawn to scale.

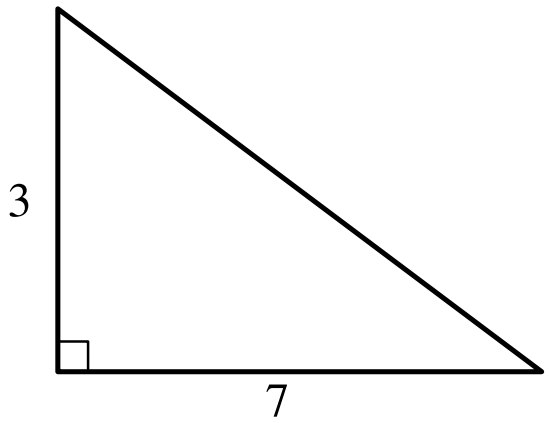
Which equation shows the relationship between the side lengths of the triangle shown?

A. $12 \cdot 3 = c$

B. $12 + 3 = c$

C. $12^2 + 3^2 = c^2$

D. $12^2 - 3^2 = c^2$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the sides adjacent to the right angle are as follows:
  - 3
  - 7
- A note indicates the figure is not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

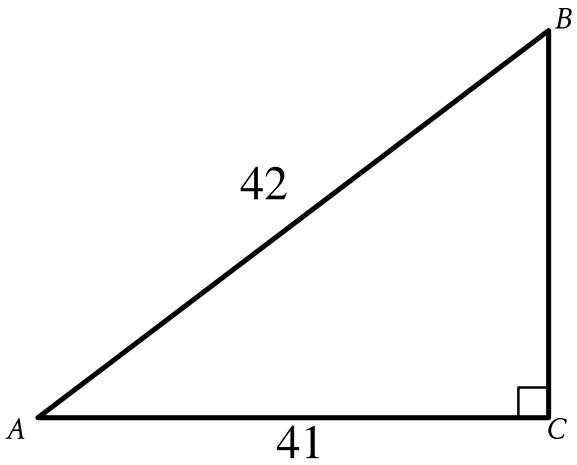
A. 3.2

B. 5

C. 7.6

D. 20

9 8d9a8a64-b22e-4435-a19b-1ac97f998d7a · Medium



Note: Figure not drawn to scale.

- Angle upper A upper C upper B is a right angle.
- The length of side upper A upper B is 42.
- The length of side upper A upper C is 41.
- A note indicates the figure is not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

A. $\frac{42}{41}$

B. $\frac{41}{42}$

C. $\frac{1}{42}$

D. $\frac{1}{41}$

---

10 aa7ee350-2124-4fd5-afe1-f10a1f785d05 · Medium

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

A. 34

B. 102

C. 136

D. 170

---

11 b479bf47-357b-4283-81e0-02fa0f919256 · Medium

A right triangle has legs with lengths of 28 centimeters and 20 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $8\sqrt{6}$

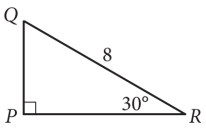
B. $4\sqrt{74}$

C. 48

D. $1{,}184$

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12 09170-DC · Medium



In the right triangle shown above, what is the length of $\overline{PQ}$?

Student-produced response: _____

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13 19da9e4a-267f-405a-bab5-768d4d5c4c29 · Medium

In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin R$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos S$?

A. $\frac{\sqrt{15}}{15}$

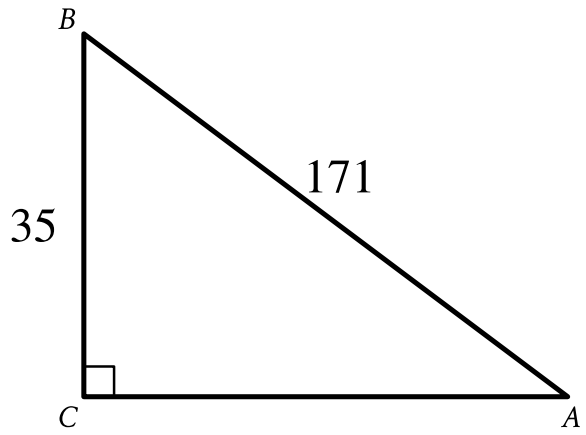
B. $\frac{\sqrt{15}}{4}$

C. $\frac{4\sqrt{15}}{15}$

D. $\sqrt{15}$

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14 8d60c337-175f-44ef-817b-67041302a3bd · Medium



Note: Figure not drawn to scale.

- Angle upper C is a right angle.
- The length of side upper A upper B is 171.
- The length of side upper B upper C is 35.
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

A. $\frac{1}{171}$

B. $\frac{35}{171}$

C. $\frac{171}{35}$

D. 171

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15 c15bdfb8-01a8-4d34-9cc6-822ca3108767 · Medium

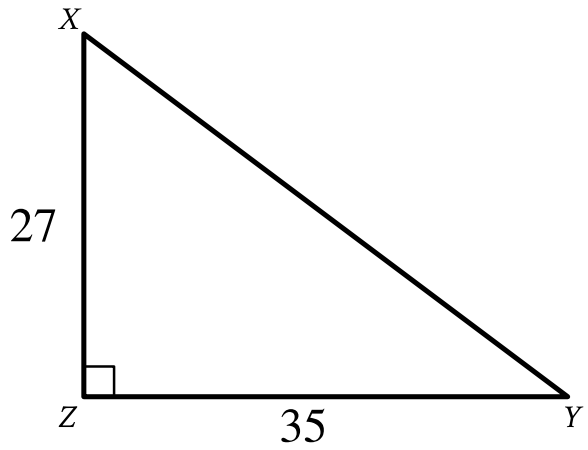
A right triangle has legs with lengths of 11 centimeters and 9 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $\sqrt{40}$

B. $\sqrt{202}$

C. 20

D. 202



Note: Figure not drawn to scale.

- Angle upper Z is a right angle.
- The length of side upper X upper Z is 27.
- The length of side upper Y upper Z is 35.
- A note indicates the figure is not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

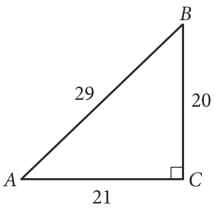
A. $\tan X$

B. $\tan Y$

C. $\cos X$

D. $\cos Y$

17 08251-DC · Medium



In the figure above, what is the value of $\tan(A)$?

A. $\frac{20}{29}$

B. $\frac{21}{29}$

C. $\frac{20}{21}$

D. $\frac{21}{20}$

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18 076bf06d-2022-480c-ada2-6ea25ff29991 · Medium

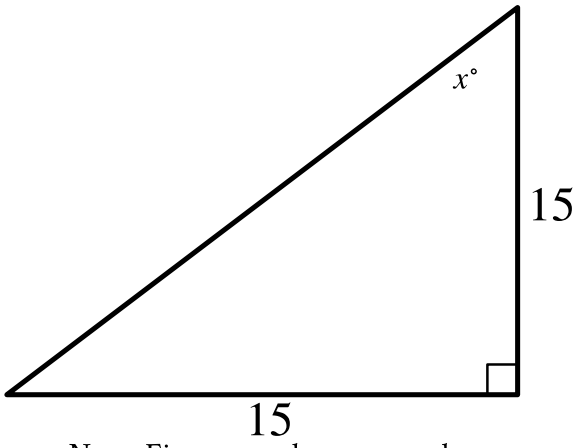
One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

A. 43.2

B. 120

C. 192

D. 201.5

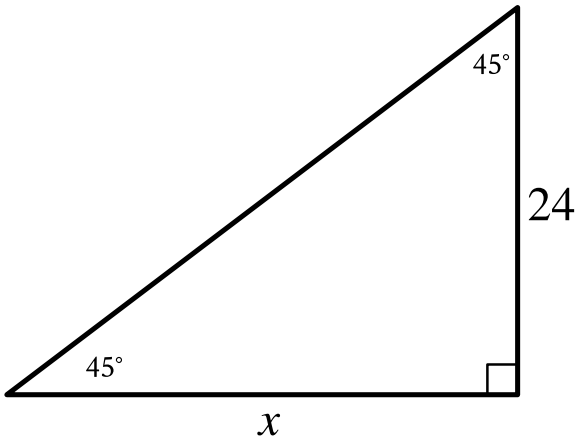


Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 15
  - 15
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x?

Student-produced response: _____



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of the other 2 angles are as follows:
  - 45°
  - 45°
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - x
  - 24
- Note: Figure not drawn to scale.

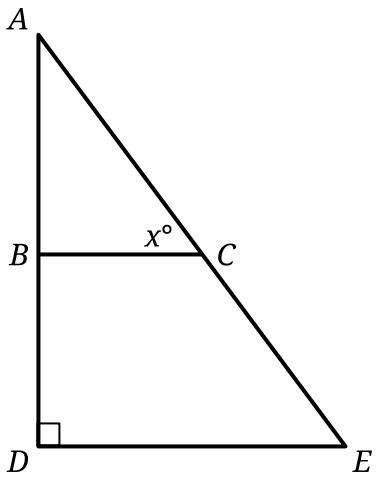
In the triangle shown, what is the value of x?

A. 24

B. 45

C. 48

D. 69



Note: Figure not drawn to scale.

- Angle upper A upper D upper E is a right angle.
- Line segment upper B upper C is drawn from side upper A upper D to side upper A upper E.
- The measure of angle upper B upper C upper A is x°.
- Note: Figure not drawn to scale.

In the figure, $\bar{BC}$ is parallel to $\bar{DE}$. If the length of $\bar{DE}$ is 147 and the length of $\bar{AE}$ is 245, what is the value of $\tan x°$?

A. $\frac{147}{196}$

B. $\frac{147}{245}$

C. $\frac{196}{147}$

D. $\frac{245}{147}$

22 ef234bbf-719d-4bc2-a841-937a532d135b · Medium

The length of a rectangle's diagonal is $5\sqrt{17}$, and the length of the rectangle's shorter side is 5. What is the length of the rectangle's longer side?

A. $\sqrt{17}$

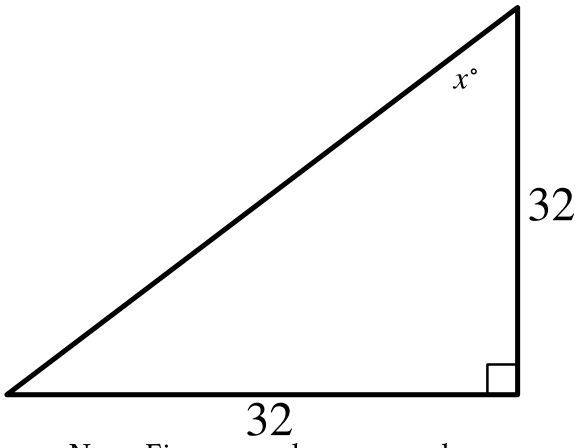
B. 20

C. $15\sqrt{2}$

D. 400

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23 c9527c06-e018-47ad-b3bf-4779309e09b7 · Medium



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 32
  - 32
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of *x*?

Student-produced response: _____

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24 62bc2dfe-0a0c-45ec-a764-c5f84ddbce67 · Medium

Triangle *FGH* is similar to triangle *JKL*, where angle *F* corresponds to angle *J* and angles *G* and *K* are right angles. If $\sin F = \frac{308}{317}$, what is the value of $\sin(J)$?

A. $\frac{75}{317}$

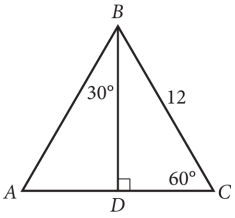
B. $\frac{308}{317}$

C. $\frac{317}{308}$

D. $\frac{317}{75}$

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25 04175-DC · Medium



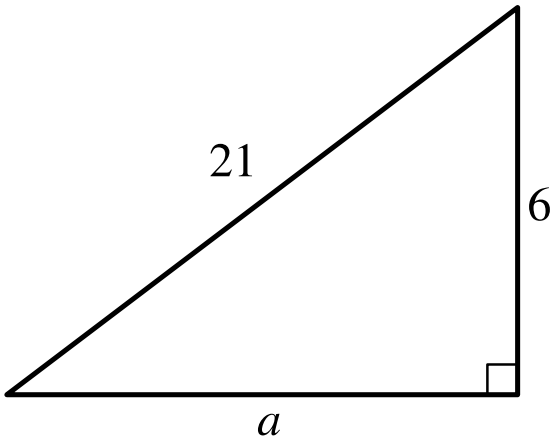
In ΔABC above, what is the length of $\overline{AD}$?

A. 4

B. 6

C. $6\sqrt{2}$

D. $6\sqrt{3}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 21.
- The length of one side adjacent to the right angle is 6.
- The length of the other side adjacent to the right angle is a.
- A note indicates the figure is not drawn to scale.

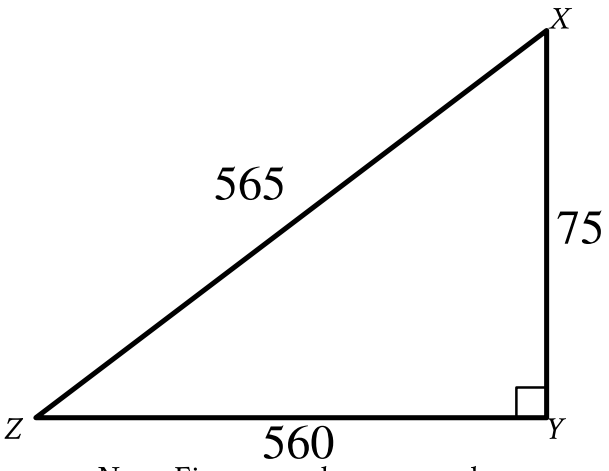
For the triangle shown, which expression represents the value of *a*?

A. $\sqrt{21^2 - 6^2}$

B. $21^2 - 6^2$

C. $\sqrt{21 - 6}$

D. $21 - 6$



Note: Figure not drawn to scale.

- Angle upper Y is a right angle.
- The length of side upper X upper Y is 75.
- The length of side upper Y upper Z is 560.
- The length of side upper Z upper X is 565.
- A note indicates the figure is not drawn to scale.

In triangle XYZ, $\cos Z = \frac{560}{c}$. What is the value of c?

Student-produced response: _____

$RS = 440$

$ST = 384$

$TR = 584$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

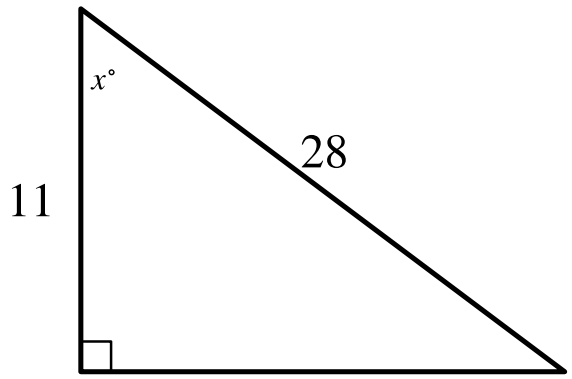
A. $\frac{48}{73}$

B. $\frac{55}{73}$

C. $\frac{48}{55}$

D. $\frac{55}{48}$

29 9312f237-66e1-4380-a28d-185ceec5d2ab · Hard



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the hypotenuse is 28.
- The length of the leg adjacent to the angle with measure x° is 11.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\cos x°$?

Student-produced response: _____

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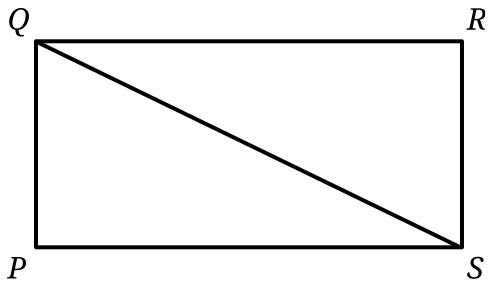
30 ddfe8011-1fec-4e7d-81bb-e842017e4525 · Hard

Triangle *ABC* is similar to triangle *DEF*, where angle *A* corresponds to angle *D* and angle *C* corresponds to angle *F*. Angles *C* and *F* are right angles. If $\tan A = \frac{50}{7}$, what is the value of $\tan E$?

Student-produced response: _____

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31 35758801-d927-43ce-8662-b8d768411b28 · Hard



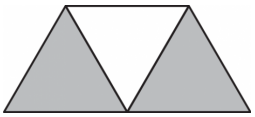
Note: Figure not drawn to scale.

In rectangle *PQRS*, the length of line segment *QS* is $\sqrt{2{,}344}$ and the length of side *PS* is 8 more than the length of side *RS*. What is the length of side *QR*?

Student-produced response: _____

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32 07822-DC · Hard



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

A. $2\sqrt{3}$

B. $4\sqrt{3}$

C. $8\sqrt{3}$

D. 16

---

33 e8c34144-2ff1-46f4-99ab-36c137fcb022 · Hard

In right triangle *QRS*, the length of hypotenuse *SQ* is $4\sqrt{82}$ and the length of side *QR* is 4. What is the length of side *RS*?

Student-produced response: _____

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34 8c6f39d2-0804-4eb5-8c97-b801dcd8a2e8 · Hard

Triangle *ABC* is similar to triangle *DEF*, where *A* corresponds to *D* and *C* corresponds to *F*. Angles *C* and *F* are right angles. If $\tan A = \sqrt{3}$ and $DF = 125$, what is the length of $\overline{DE}$?

A. $125\frac{\sqrt{3}}{3}$

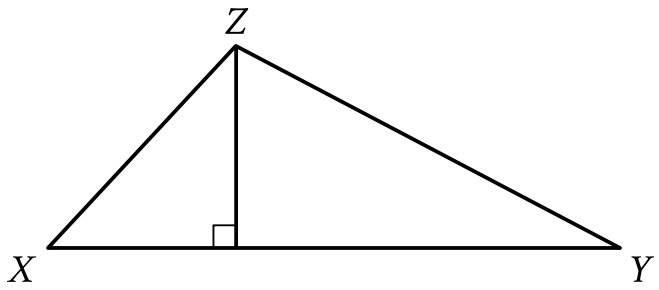
B. $125\frac{\sqrt{3}}{2}$

C. $125\sqrt{3}$

D. 250

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35 d8a83c5c-3c86-4caa-91c0-283aa8c5d6b0 · Hard



Note: Figure not drawn to scale.

- A line segment is drawn from vertex upper Z to side upper X upper Y.
- The angle at the intersection of the line segment and side upper X upper Y is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, the measure of angle X is $54°$. The length of XY is 26 units and the length of XZ is 19 units. What is the area, in square units, of triangle XYZ?

A. 247

B. 494

C. $247\sin 54°$

D. $494\sin 54°$

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36 575135d8-2a82-4346-993f-f5d9b5921696 · Hard

The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3. What is the length of the rectangle's longer side?

Student-produced response: _____

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37 889dae43-01c0-449b-abd7-1a385a85c6a7 · Hard

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x?

A. 5

B. 19

C. 23

D. 29

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38 5bf76a41-5cb0-4d02-8d3a-4c2fc8668b3a · Hard

Which of the following expressions is equivalent to $\sin 24° \cos 66° + \cos 24° \sin 66°$?

A. $2\cos 66° \sin 24°$

B. $2\cos 66° + 2\cos 24°$

C. $\cos 66°^2 + \cos 24°^2$

D. $\cos 66°^2 + \sin 24°^2$

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39 09173-DC · Hard

A triangle with angle measures 30°, 60°, and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

Student-produced response: _____

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40 a3d7bd43-4b60-4b1e-a039-96f3769a34f2 · Hard

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. What is the value of x?

A. 5

B. 13

C. 23

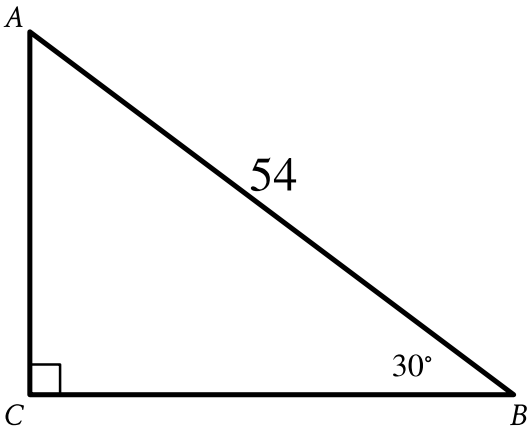
D. 38

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41 4363b40f-7867-454b-a423-9aa19a6be480 · Hard

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1{,}089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

Student-produced response: _____



Note: Figure not drawn to scale.

- The length of side upper A upper B is 54.
- The measure of angle upper B is 30°.
- Angle upper C is a right angle.
- A note indicates the figure is not drawn to scale.

Right triangle *ABC* is shown. What is the value of $\tan A$?

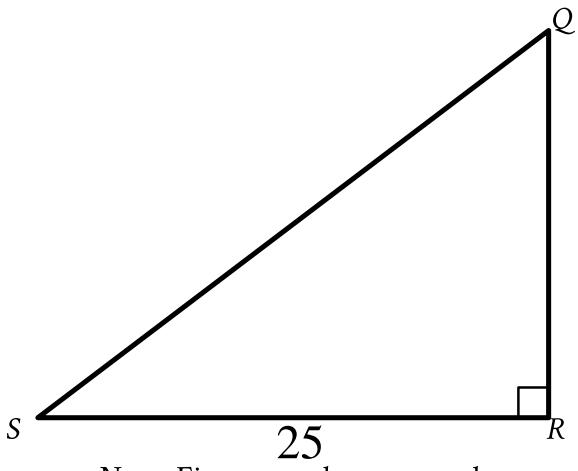
A. $\frac{\sqrt{3}}{54}$

B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. $27\sqrt{3}$

43 383c05a2-4e2a-4c4e-8121-840ca478825a · Hard



Note: Figure not drawn to scale.

- Angle upper R is a right angle.
- The length of side upper R upper S is 25.
- A note indicates the figure is not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A. $25\cos Q$

B. $25\sin Q$

C. $\frac{25}{\cos Q}$

D. $\frac{25}{\sin Q}$

---

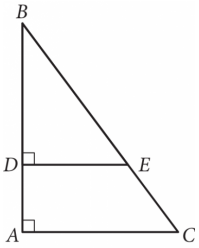
44 081f3aa6-88ed-4da7-bc3a-e7bbfa7b4fca · Hard

The perimeter of an equilateral triangle is 852 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?

Student-produced response: _____

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45 00615-DC · Hard



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

Student-produced response: _____

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46 af8e7cb1-d233-45ae-8811-379283e1e3a4 · Hard

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

A. 9

B. $9\sqrt{2}$

C. 18

D. $18\sqrt{2}$

---

47 e8f7d348-9106-4f42-a41c-8400c1758a07 · Hard

In triangle XYZ, angle Y is a right angle, the measure of angle Z is $33°$, and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k\tan 33°$, where k is a constant, what is the value of k?

Student-produced response: _____

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48 028928-DC · Hard

In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

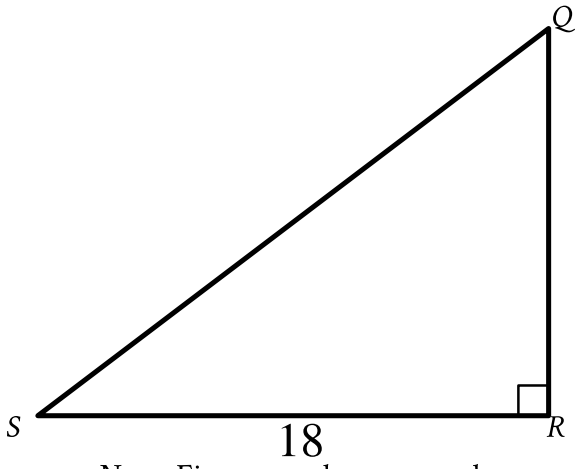
A. $-\frac{\sqrt{3}}{3}$

B. $-\frac{3}{\sqrt{3}}$

C. $\frac{\sqrt{3}}{3}$

D. $\frac{3}{\sqrt{3}}$

49 60ba6bab-7d16-46f1-9882-e80e2dd3577c · Hard



Note: Figure not drawn to scale.

- Angle upper R is a right angle.
- The length of side upper R upper S is 18.
- A note indicates the figure is not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A. $18\cos Q$

B. $18\sin Q$

C. $\frac{18}{\cos Q}$

D. $\frac{18}{\sin Q}$

---

50 7dae086f-c99c-48f6-b4ad-68e5e1008035 · Hard

In right triangle XYZ, angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is $a°$, the measure of $\angle XYR$ is $b°$, and the length of $\overline{XY}$ is 19. If $\sin a° = \frac{12}{13}$, what is the value of $\tan b°$?

Student-produced response: _____

---

51 ccc97dea-b103-4868-86e4-80b756cd63d7 · Hard

In triangle *ABC*, angle *B* is a right angle. The length of side *AB* is $10\sqrt{37}$ and the length of side *BC* is $24\sqrt{37}$. What is the length of side *AC*?

A. $14\sqrt{37}$

B. $26\sqrt{37}$

C. $34\sqrt{37}$

D. $\sqrt{34 \cdot 37}$

---

52 16428dd4-6b32-4580-ace2-d2dbf0505a76 · Hard

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

A. 47

B. $47\sqrt{2}$

C. 94

D. $94\sqrt{2}$

---

53 8ee41a37-b82e-4974-ab8e-ab0531faca66 · Hard

In triangle *XYZ*, the measure of angle *X* is $90°$. Point *W* lies on segment *YZ*, and segment *WX* is perpendicular to segment *YZ*. The length of segment *WY* is 572, and the length of segment *WX* is 429. What is the value of $\tan Z$?

A. $\frac{3}{5}$

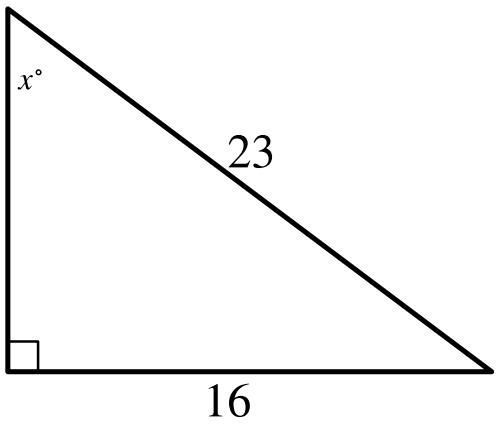
B. $\frac{3}{4}$

C. $\frac{4}{5}$

D. $\frac{4}{3}$

---

54 c66e94ab-e172-4e0b-9609-f3b62ef3f90d · Hard



Note: Figure not drawn to scale.

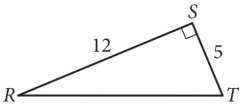
- One angle is a right angle.
- The measure of a second angle is x°.
- The length of the hypotenuse is 23.
- The length of the leg opposite the angle with measure x° is 16.
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x°$?

Student-produced response: _____

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55 05777-DC · Hard



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

Student-produced response: _____

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56 13cc44bf-f2ac-4f02-ac05-440e4b894f96 · Hard

In triangle JKL, $\cos K = \frac{24}{51}$ and angle J is a right angle. What is the value of $\cos L$?

Student-produced response: _____

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57 1dd30616-5d7f-4401-8dd7-811d477d90d8 · Hard

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k?

Student-produced response: _____

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58 2124991b-31a9-4ce3-a963-094142d78f70 · Hard

In triangle XYZ, angle Z is a right angle and the length of $\overline{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ?

A. 188

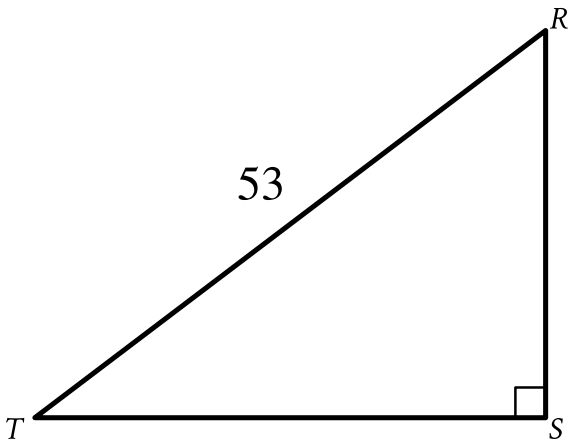
B. 168

C. 84

D. 71

---

59 3e66c309-612d-4f4e-90d5-3132cd7909e6 · Hard



Note: Figure not drawn to scale.

- Angle upper S is a right angle.
- The length of side upper R upper T is 53.
- A note indicates the figure is not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

Student-produced response: _____

---

60 fa02652e-89a4-4784-990e-fd5f06d65862 · Hard

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

A. 20

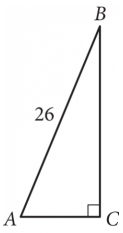
B. $\frac{20\sqrt{2}}{2}$

C. $20\sqrt{2}$

D. 40

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61 029664-DC · Hard

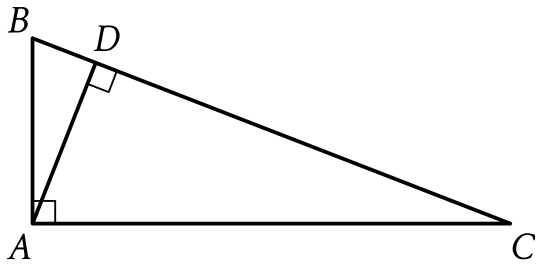


Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Student-produced response: _____

---

62 988fed36-4761-4d7b-962b-35933cf1b89e · Hard



Note: Figure not drawn to scale.

- Angle upper B upper A upper C is a right angle.
- Point upper D is on side upper B upper C.
- Line segment upper A upper D is drawn from vertex upper A to side upper B upper C.
- Angle upper A upper D upper C is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure, $\bar{AD}$ and $\bar{BC}$ intersect at point D. If the length of $\bar{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\bar{AD}$?

Student-produced response: _____

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63 2ec14ac8-bf5b-46cc-8993-b907b006b31c · Hard

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angles C and F are right angles. The length of $\bar{AB}$ is 2.9 times the length of $\bar{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

Student-produced response: _____

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64 84b26634-2244-45e6-8504-ea367a0f2049 · Hard

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $\frac{5}{13}$

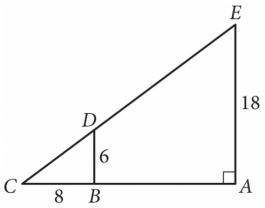
B. $\frac{5}{12}$

C. $\frac{12}{13}$

D. $\frac{12}{5}$

---

65 05758-DC · Hard



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

Student-produced response: _____

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66 1ef8e9cb-4e5c-441a-9395-222ae1a20a61 · Hard

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t?

A. $6t + \pi$

B. $6t$

C. $5t + \pi$

D. $5t + \frac{\pi}{2}$

---

67 ae6da77a-4de7-4375-a532-b00350663a9f · Hard

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d?

Student-produced response: _____

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68 5e8901b5-d7f3-45e7-8324-de2b20574706 · Hard

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

A. 117

B. $117\sqrt{3}$

C. 234

D. $1{,}521\sqrt{3}$

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69 4959ea4c-8faf-4a7c-9ef9-60dc6d45d751 · Hard

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

A. $29\sqrt{2}$

B. $58\sqrt{2}$

C. $58 + 58\sqrt{2}$

D. $58 + 116\sqrt{2}$